

Operators Activity #2

PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

CODE

```
public class OperatorsPart1Num1 {  
    Run | Debug  
    public static void main(String[] args){  
  
        boolean a = true & true;  
        boolean b = true ^ true;  
  
        int c = 8 | 2;  
        int d = 10 | 5;  
  
        System.out.println(a + ", " + c + ", " + b + ", " + d);  
    }  
}
```

OUTPUT

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```
C:\Users\Administrator\Downloads\OOP Class Codes> cmd /C ""C:\Program Files\Eclipse Adoptium\jdk-17.0.15.6-hotspot\bin\java.exe" -XX:+ShowCodeDetailsInExceptionMessages -cp "C:\Users\Administrator\AppData\Roaming\Code\User\workspaceStorage\6adba4fe068345416f156bd7677d4139\redhat.java\jdt_ws\OOP Class Codes_58d44e89\bIn" OperatorsPart1Num1 "  
true, 10, false, 15
```

2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

CODE

```

public class OperatorsPart1Num2{
    Run | Debug
    public static void main(String[] args) {
        boolean a = (10 > 3)? true : false;
        boolean b = (10 < 3)? true : false;

        int num1 = (10 > 5)? 25 : 10;
        int num2 = (10 < 5)? 11 : 30;

        System.out.println(a + ", " + b + ", " + num1 + ", " + num2);
    }
}

```

OUTPUT

```

C:\Users\Administrator\Downloads\OOP Class Codes> cmd /C ""C:\Pr
DetailsInExceptionMessages -cp "C:\Users\Administrator\AppData\Roaming\
\jdt_ws\OOP Class Codes_58d44e89\bin" OperatorsPart1Num2 "
true, false, 25, 30

```

3. Create a Java program using assignment operators that will produce the output:
Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

CODE

```

public class OperatorsPart1Num3{
    Run | Debug
    public static void main(String[] args) {
        int result = 90;

        result +=30;
        int a = result;

        result -=50;
        int b = result;

        result = 12;
        result*=12;
        int c = result;

        result/=9;
        int d = result;

        System.out.println("Addition: " + a + ", Subtraction : " + b + ", Multiplication : " + c + ", Division : " + d );
    }
}

```

OUTPUT

```
C:\Users\Administrator\Downloads\OOP Class Codes> cmd /C ""C:\Program Files\E
DetailsInExceptionMessages -cp "C:\Users\Administrator\AppData\Roaming\Code\U
\jdt_ws\OOP Class Codes_58d44e89\bin" OperatorsPart1Num3 "
Addition: 120, Subtraction : 70, Multiplication : 144, Division : 16
```

PART II:

Program Requirements:

1. Use at least five (5) different operators from the following categories:
 - Arithmetic operators (+, -, *, /, %)
 - Relational operators (>, <, >=, <=, ==, !=)
 - Logical operators (&&, ||, !)
 - Unary operators (++ , --)
 - Assignment operators (+ =, - =, * =, / =)
 - Bitwise operators (&, |, ^, <<, >>)
2. The program must produce exactly six (6) outputs only.
3. At least one output must include a tricky increment or decrement case, such as:
 - `a++ + ++a`
 - `--b + b++`
 - or any similar expression that modifies the variable within the same statement.
4. Do NOT reuse:
 - Any variable values
 - Any variable names
 - Any logic patterns from Part I.

5. The output must contain:

- At least two (2) boolean results
- At least four (4) numeric results

CODE

```
public class OperatorsPart2 {  
    Run | Debug  
    public static void main(String[] args) {  
        int a = 49;  
        int b = 11;  
  
        boolean res1 = (b > a)? true : false;  
        boolean res2 = ((a & b) >= 10)? true: false;  
        System.out.println(res1);  
        System.out.println(res2);  
        System.out.println(a++ - ++b + ++a);  
        System.out.println(b*=3);  
        System.out.println(b-- + ++a - --a);  
        System.out.println(a % b);  
    }  
}
```

OUTPUT

```
false  
true  
88  
36  
37  
16
```